

6.3 SAMPLE CODE

6.3.1 ANDROID WEBVIEW INTEGRATION FOR UPI FRAUD DETECTION APP

```
J ExampleInstrumentedTest.java ×
app > src > androidTest > java > com > example > upi > J ExampleInstrumentedTest.java
1  package com.example.upi;
2
3  import android.content.Context;
4
5  import androidx.test.platform.app.InstrumentationRegistry;
6  import androidx.test.ext.junit.runners.AndroidJUnit4;
7
8  import org.junit.Test;
9  import org.junit.runner.RunWith;
10
11 import static org.junit.Assert.*;
12
13 /**
14  * Instrumented test, which will execute on an Android device.
15  *
16  * @see <a href="http://d.android.com/tools/testing">Testing documentation</a>
17  */
18 @RunWith(AndroidJUnit4.class)
19 public class ExampleInstrumentedTest {
20     @Test
21     public void useAppContext() {
22         // Context of the app under test.
23         Context appContext = InstrumentationRegistry.getInstrumentation().getTargetContext();
24         assertEquals("com.example.upi", appContext.getPackageName());
25     }
26 }
```

Fig 6.3.1 Integration code

6.3.2 SIMPLE UNIT TEST FOR ANDROID APPLICATION

```
J ExampleUnitTest.java ×
app > src > test > java > com > example > upi > J ExampleUnitTest.java
1  package com.example.upi;
2
3  import org.junit.Test;
4
5  import static org.junit.Assert.*;
6
7  /**
8  * Example local unit test, which will execute on the development machine (host).
9  *
10 * @see <a href="http://d.android.com/tools/testing">Testing documentation</a>
11 */
12 public class ExampleUnitTest {
13     @Test
14     public void addition_isCorrect() {
15         assertEquals(4, 2 + 2);
16     }
17 }
```

Fig 6.3.2 Unit Test Code

6.3.3 ANDROID WEBVIEW IMPLEMENTATION FOR LOADING UPI FRAUD DETECTION WEBSITE

```

J MainActivity.java X
app > src > main > java > com > example > upi > J MainActivity.java
1  package com.example.upi;
2
3  import androidx.appcompat.app.AppCompatActivity;
4
5  import android.os.Bundle;
6  import android.webkit.WebSettings;
7  import android.webkit.WebView;
8  import android.webkit.WebViewClient;
9
10 public class MainActivity extends AppCompatActivity {
11
12     @Override
13     protected void onCreate(Bundle savedInstanceState) {
14         super.onCreate(savedInstanceState);
15         setContentView(R.layout.activity_main);
16
17         WebView webView = findViewById(R.id.webView);
18         webView.setWebViewClient(new WebViewClient()); // Ensures the WebView handles URLs
19         WebSettings webSettings = webView.getSettings();
20         webSettings.setJavaScriptEnabled(true); // Enable JavaScript
21         webView.loadUrl("http://172.20.200.56:5000/");
22     }
23 }

```

Fig 6.3.3 Upload Code

6.3.4 CNN FILE FOR WEBSITE

```

cnn.py
1  import pandas as pd
2  import numpy as np
3  import matplotlib.pyplot as plt
4  import seaborn as sns
5  import pandas as pd
6
7  file_name = "UPI_Fraud_Dataset_Expanded.csv"
8  df = pd.read_csv(file_name)
9
10 x = df.iloc[:, :8].values
11 y = df.iloc[:, 8].values
12
13 from sklearn.model_selection import train_test_split
14 x_train, x_test, y_train, y_test = train_test_split(x, y, test_size=0.3, random_state=42)
15
16 x_test.shape
17 x_train.shape
18
19 y_test.shape
20 y_train.shape
21
22 from sklearn.tree import DecisionTreeClassifier
23 from sklearn.metrics import classification_report, accuracy_score, precision_score, recall_score, f1_score, confusion_matrix
24 import matplotlib.pyplot as plt
25 import seaborn as sns
26
27 # Create a DecisionTreeClassifier instance
28 decision_tree = DecisionTreeClassifier(random_state=42)
29
30 # Train the decision tree model
31 decision_tree.fit(x_train, y_train)
32
33 # Predict on the test set
34 y_pred = decision_tree.predict(x_test)
35
36 # Calculate accuracy
37 accuracy = accuracy_score(y_test, y_pred)
38
39

```

Fig 6.3.4 CNN File